Production electrical and functional test procedure

USBGE-TST-001 | Document revision 04 | 2026-10-16

Design baseline USBGE-A-WT-20261016-R04 | A (native schematic title revision A0)

Execution and limits

This is an executable test instruction with unexecuted record templates. It contains established device operating bounds and explicitly identified engineering screen proposals. No board was powered or physically tested during package preparation. Production dispatch requires approved fixture/current/traffic limits and first-article qualification under RR-07.

Execute PT-01 through PT-10 in order. Stop at a fail or missing mandatory limit, remove power if appropriate, label HOLD and raise an NCR. Do not bypass a failed step by manually entering PASS. Retest affected and regression steps after authorised rework.

Use GND candidate C1.1 as the meter reference; rail pads and XY coordinates are in USBGE-DFT-002. Verify the intended ground physically before energising. Never use shield GNDU/GNDE as an assumed ground. The first-article engineer must prove probe access and fixture repeatability.

Equipment and setup

ID	Equipment	Qualification requirement
EQ-01	Qualified USB host	SuperSpeed and separate USB2 path; fixed OS/driver image; record host ID and port topology.
EQ-02	USB power fixture / meter	Data-isolated current-limited 5 V supply, inline current logging; prohibit host VBUS backfeed; verify fixture wiring before connection.
EQ-03	DMM and microprobes	Calibrated rail-voltage / low-energy resistance measurement; record uncertainty, range and probe location.
EQ-04	Oscilloscope	Suitable differential/low-inductance probing for engineering rail/ripple diagnosis; no USB compliance conclusion from ordinary scope traces.
EQ-05	Ethernet partner and cables	Known-good 10/100/1000 capable partner, Cat5e or better cable; qualified host baseline avoids bottleneck misdiagnosis.
EQ-06	Inspection equipment	Magnification, AOI/SPI where qualified, hidden-joint X-ray; dimensional gauges.
EQ-07	Test software	OS USB topology/descriptors and NIC tools, iperf3 and hash utility; fixed versions and command logs.
EQ-08	ESD / traceability equipment	ESD workstation, serial label equipment and controlled traveler database or signed forms.

Establish a segregated local test network. Assign host and partner addresses in the approved test configuration; disable routing/bridging to other networks during traffic tests. Verify the host and partner using a known-good adapter before judging throughput. Store all results under the serial and station-run ID.

Sequential procedure

Step / test	Operation	Acceptance / disposition
PT-01 / Document / lot verification	Identify board and BOM revision; check PCB lot, assembly traveler, material lots, equipment calibration and test-software revision.	Exact approved release manifest and unique serial linked to traveler; otherwise HOLD.
PT-02 / Incoming bare PCB	Inspect contour, board damage, layer/mask registration, holes and solderable surfaces under magnification. Measure length, width and thickness; review supplier electrical-test and impedance-coupon reports.	70 x 34 mm nominal contour; thickness/tolerances and IPC class must match the agreed purchase specification. No conductor opens/shorts or visible damage. Missing contractual limits => HOLD, not PASS.

Step / test	Operation	Acceptance / disposition
PT-03 / PCBA visual / hidden joints	Inspect all 48 references for correct values/identity, missing/extra parts, orientation, solder bridges, tombstones, lifted pads, contamination and connector fit. Inspect U3 exposed pad and leadless hidden joints by qualified X-ray.	Population matches BOM; pin 1 and connector orientation agree with numbered-pad drawing. Workmanship meets the contract-selected IPC-A-610/J-STD-001 criteria; no unassessed hidden joints.
PT-04 / Unpowered electrical screen	Disconnect all cables, discharge capacitors, verify GND continuity and rail-to-ground resistance using low-energy DMM. Compare with an approved known-good first article; log range, polarity and settling time.	No metallic short, damaged pad or foreign conductor. Rail resistance threshold requires first-article qualification; semiconductor/charging readings are not a universal ohmic pass criterion. Outlier => HOLD.
PT-05 / Controlled power-up	Use an isolated, current-limited USB power fixture at 5.00 V setpoint with host data disconnected. Establish current limit from approved first-article plan before energising. Monitor input current and rails immediately; remove power for current limit, abnormal heating or rail excursions.	No current-limit event or abnormal heating. +3V3 2.97..3.63 V; +2V5 2.38..2.63 V; +1V2/+1V2A 1.10..1.26 V at the identified pads. Use measurement uncertainty guard bands. These are device operating bounds, not a qualified process-control window. Input current/ripple limits require qualification.
PT-06 / USB SuperSpeed / fallback	Connect known-good SuperSpeed Type-A-to-Type-B cable to a qualified host. Record descriptors, VID/PID, serial/config readback and negotiated speed. Repeat on a USB2-only path. Power-cycle and reconnect five times per path under the proposed qualification plan.	Correct approved identity; USB SuperSpeed 5000 Mbit/s on qualified SS path and 480 Mbit/s on USB2 path. No unexpected disconnect, enumeration error or driver fault. Cycle count is an engineering test-plan proposal requiring qualification, not an existing product requirement.
PT-07 / Ethernet link and integrity	Use Cat5e-or-better cable and qualified partner; exercise auto-negotiation at 1000 full duplex, and constrained 100/10 modes. Record link state, LED behavior and counters. Perform 100 ICMP echoes on isolated static-IP network.	Expected negotiated rates; 1000BASE-T full duplex. Zero lost echoes in the controlled test. LED mapping must match approved configuration. Any unexplained CRC/error-counter increase => HOLD. Ping count is a proposed screen, not IEEE conformance evidence.
PT-08 / Bidirectional data exercise	Run checksum-verified file transfer in both directions; run iperf3 TCP for 60 s each direction at 1 Gbit/s link. Save command, versions, byte counts, throughput, retransmissions and before/after NIC counters.	File checksums equal; no link reset or device disconnect. Throughput minimum and retransmission limit must be established with qualified host/cable/partner; until entered into the approved plan this step cannot release units. Do not equate 1 Gbit/s link rate with payload throughput.
PT-09 / Identity and configuration	Read MAC and USB identity after a cold reconnect. Compare approved image hash, MAC allocation and expected descriptors. Program OTP only under separately approved executable work instruction.	Unique approved MAC/identity persists and readback matches. No OTP write is authorised by this document. Blank/duplicate/temporary identities => HOLD.
PT-10 / Final disposition / packing	Review every record and attachment, resolve NCRs and rework retest, apply approved serial label clear of contacts/vents, package in ESD protection and record shipping lot.	All mandatory steps PASS with signed reviewer disposition and correct release revision; missing limits, skipped tests or open NCRs prevent dispatch.

Software execution record

On Linux, example read-only diagnostics are lsusb -t (USB speed), lsusb -v (descriptors), ip link (interface identity), ethtool <interface> (link state) and ethtool -S <interface> (counters). Record interface name and software version; these commands have not been run against hardware here.

Run iperf3 -s on the isolated partner. On the test host use iperf3 -c <partner-address> -t 60 --json, then iperf3 -c <partner-address> -t 60 -R --json. Save JSON and both peers' counter snapshots. Transfer a known test file in each direction and compare SHA-256 at source and destination. The placeholders identify station settings, not missing test steps.

Device-specific descriptor/OTP readback needs an approved configuration image and tool; do not guess VID/PID or write irreversible OTP. Test procedure release includes locking software versions, required privileges, network addresses and expected identity.

Acceptance basis

LAN7800 operating rail bounds are from Microchip DS00001992J section 16.2, page 270. Record actual voltages and instrument uncertainty; passing an operating bound alone does not establish good ripple, sequencing, lifetime margin or USB current compliance. Nominal USB fixture voltage is a test setup proposal.

USB and Ethernet negotiated rates are functional observations, not certification results. Electrical USB compliance, Ethernet PHY compliance, isolation/hi-pot, EMC/ESD immunity, environmental and product safety qualification are outside this local execution. Any such contractual requirement must be validated by the appropriate qualified test plan.